Spec-Driven Artificial Intelligence for Empirical Research: A Scoping Review and an Architecture of Epistemic Control

Grzegorz Wojarnik

July 2026

University of Szczecin, Institute of Economics and Finance, Szczecin, Poland
ORCID: 0000-0001-6946-547X · e-mail: grzegorz.wojarnik@usz.edu.pl

Abstract

AI-assisted research systems increasingly automate literature search, coding, statistical analysis, visualization, and manuscript drafting. Automation alone, however, does not resolve the underlying methodological problem of empirical research: how to distinguish exploration from confirmation, preserve provenance, control the latitude of analytical decisions, and ensure that claims do not outrun the evidentiary status of the data and the study design. This article is a scoping review of six approaches bearing on this challenge - preregistration and registered reports, pre-analysis plans in economics, multiverse analysis and specification curve, reproducible research and research compendia, FAIR and provenance standards, and emerging AI-for-science workflows. We synthesize these approaches into a spec-driven AI methodology for empirical research (FSAIM), announced separately as a position note [10.2139/ssrn.6853581], in which hypotheses, data, code, artifacts, claims, and AI actions are treated as auditable objects within a staged workflow featuring human gates for approval. The proposed framework casts AI not as a shortcut for writing but as a component of an epistemically bounded research engine. We discuss implications for empirical economics and finance, where secondary data, look-ahead bias, specification multiplicity, and publication incentives create persistent risks to credibility.

---

1. Introduction

Artificial intelligence today accelerates every stage of empirical research: it finds the literature, writes the code, runs the analysis, generates the figures, and drafts the text. This acceleration, however, carries a quiet cost. Automation does not remove the fundamental vulnerability of empirical research - the analytical flexibility whereby showing only the one combination of decisions that “worked” yields a result that looks robust but is, in large part, accidental. Worse, AI adds new risks on top of this, risks that existing safeguards were never designed to catch: fabricated citations and results, hidden decisions buried in prompts, model drift across API versions, and a spurious form of audit in which a second model “confirms” the output of the first while sharing its very same errors. Research conducted with AI can therefore be simultaneously faster and less trustworthy whenever the AI’s involvement remains undocumented.

Science is not defenseless against this - it has developed a range of safeguards: preregistration and pre-analysis plans, multiverse analysis, code and data reproducibility, FAIR standards and provenance practices, and journal data/code policies. Each of these, however, protects only one fragment of the process, and each developed separately, in different disciplines that rarely read one another. No coherent framework binds them into a single workflow, and none of them covers AI accountability, since they all predate the era of generative assistants. This article addresses that gap.

The contribution is twofold. First, a map: a scoping review of the scattered credibility-control literature, organized into six approaches (layers L1-L6). Second, a framework: a synthesis of these approaches into spec-driven development (SDD), understood as an architecture of epistemic control. The central thesis is stated plainly: SDD is best understood not as a tool for writing or coding with AI, but as a way of working that freezes intentions, data, code, results, and AI involvement as auditable objects arranged into stages with human-approval gates. On this view, SDD is a compiler of good practice - it assembles the scattered approaches into a single executable workflow and adds the one element missing from all of them: AI accountability. Answers to these questions make a difference for three audiences - empirical researchers in economics and finance, editors setting data/code policy, and builders of AI-for-science tools - and for each the framework indicates a different, concrete point of entry. The framework itself carries `provisional` status: it has been tested on the single instance constituted by the production of this text, but not empirically validated against alternatives, and the full list of limitations is given in Chapter 8.

The article is structured around five research questions. Chapter 2 establishes the need for workflow-level control, with particular attention to economics and finance. Chapter 3 describes the review method. Chapter 4 maps the six approaches (RQ1: which approaches solve which fragments of the problem). Chapter 5 synthesizes them, identifying their common mechanism (RQ2) and what they jointly fail to cover (RQ3). Chapter 6 sets out the SDD framework and the safeguards that an AI-enabled world makes necessary (RQ4). Chapter 7 carries the framework into economics and finance (RQ5). Chapter 8 discusses risks and open questions, and Chapter 9 closes the argument and indicates the direction for validation. In keeping with the thesis, no claim made in this introduction runs ahead of what the subsequent chapters demonstrate - the introduction is a promise that the rest of the article pays off.

2. Background: why empirical research needs workflow-level control

Between a research question and a published result, a researcher makes hundreds of small decisions: which observations to exclude, how to define variables, which model specification to adopt, when to stop collecting data. Each of these, taken alone, may be defensible; the problem arises when they are chosen after seeing the results, and only the combination that confirms the expectation gets reported. Four concepts describe this mechanism from different angles, and all are anchored in this review's corpus. Kerr (1998) coined the term HARKing for formulating a hypothesis after learning the outcome and presenting it as if it had been predicted in advance - a move that inverts the logic of testing, since a hypothesis fitted to the data will always look confirmed [10.1207/s15327957pspr0203_4]. Simmons, Nelson, and Simonsohn (2011) showed experimentally that undisclosed flexibility in analytic decisions allows almost any effect to be presented as statistically significant, a phenomenon they named "researcher degrees of freedom" [10.1177/0956797611417632]. Even absent any conscious intent to deceive, the mere multiplicity of equally defensible analytic paths inflates the false-discovery rate: Steegen, Tuerlinckx, Gelman, and Vanpaemel (2016) responded with multiverse analysis, which reports the outcome across all reasonable combinations of decisions rather than a single chosen one [10.1177/1745691616658637], and Simonsohn, Simmons, and Nelson (2020) formalized this as the specification curve [10.1038/s41562-020-0912-z]. Ioannidis (2005) captured the scale of the problem, arguing that given typical test power and the pressure toward positive findings, most published results may in fact be false [10.1371/journal.pmed.0020124] - a claim the Open Science

Collaboration (2015) confirmed empirically, failing to reproduce a substantial share of effects in a mass replication sample [10.1126/science.aac4716].

The common denominator across these phenomena is not bad faith but the absence of freezing decisions in time. As long as the analysis plan is formed together with the result, no single after-the-fact safeguard can distinguish discovery from artifact - and this is precisely the rationale for workflow-level control, one that ties decisions to the moment they are made.

In economics and finance, analytic flexibility acquires an additional dimension, because the data are most often secondary - market data, macroeconomic series, panel data from vendors. The researcher does not collect the data personally but processes it, so the degrees of freedom shift from experimental design onto data handling. White (2000) showed formally how searching across many strategies on the same time series generates spuriously significant results - a problem especially acute in finance, where thousands of trading rules are routinely tested [10.1111/1468-0262.00152]. Christensen and Miguel (2018) documented the scale of this problem in empirical economics and the fragmented state of the responses to it [10.1257/jel.20171350], while Vilhuber (2020) described the state of reproducibility in economics and the role played by journals' data/code policies [10.1162/99608f92.4f6b9e67]; together, these two accounts confirm that the discipline has a mature awareness of the problem, yet its safeguards remain scattered. Financial and macroeconomic data, moreover, carry their own degrees of freedom that have no counterpart in experimental research - look-ahead bias, survivorship bias, revisions to macro series, divergent vendor definitions, choice of sample window, and multiplicity of specifications. Table 1 sets out six such types, assigning to each the corresponding freeze within the SDD frame; each of them is a silent decision about input data that standard data/code policies - which check reproducibility, not pre-specification - let pass through, and which the frame instead addresses at the data-freeze stage (elaborated in Chapter 7).

Table 1. Types of researcher degrees of freedom in financial data and their SDD freezes.

RDF type	Description	SDD freeze
Look-ahead bias	Using information unavailable at decision time (back-adjusted data, later-published indicators)	point-in-time snapshot
Survivorship bias	Sample limited to survivors (databases drop delisted funds/firms)	universe declaration + delisted entities
Data revisions	Macro series (GDP, employment) revised after release - final ≠ real-time data	vintage version (real-time DB)
Vendor definitions	Same indicator differs across vendors (Bloomberg/Refinitiv/CRSP) and versions	vendor + version + extraction date
Market timing / sample window	Choice of sample window (start/end, crisis exclusion) changes the result	sample-window prespecification in PAP
Multiplicity of specifications	Choice of model, controls, frequency among many equally defensible ones	specification curve / family declaration

Source: own elaboration.

3. Review method

The conceptual framework presented in the previous chapter - six approaches of credibility control and the outline of layers meant to organize them - did not emerge from intuition but from a systematic review of the literature. This chapter describes how that review was conducted: which type of review was chosen, which databases were used, how studies were searched, screened, and coded, and which methodological limitations should be borne in mind when interpreting the results that follow.

3.1. Review type and research questions

A scoping review method was applied, following the PRISMA-ScR framework (Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews). This choice, rather than a classical systematic review, follows from the nature of the research questions: the aim is not to synthesize a quantitative effect but to map and compare dispersed approaches of research credibility control and to identify the gaps that the proposed framework is intended to fill. A scoping review is the appropriate method when the aim is to map a heterogeneous field dispersed across disciplines and to identify gaps, rather than to estimate a pooled effect, which would require a common outcome measure. The review addresses five research questions (RQ1-RQ5) frozen in scope version 1.0, whose full wording is given in Appendix A.

The review was deliberately not conducted as a meta-analysis or as a systematic review aimed at grading the strength of evidence, because meta-analysis requires a set of studies measuring the same, comparable effect through a shared outcome measure (effect size) that can be aggregated into a pooled statistic - and the corpus of this review is heterogeneous by design. It spans six distinct approaches of credibility control - from preregistration, through multiverse analysis and reproducibility, to provenance standards, data/code policies, and AI agent accountability - which share no common effect measure, and the underlying studies do not test a common hypothesis but describe different mechanisms safeguarding different segments of the research process. Quantitative aggregation here would be not merely undesirable but without an object: no single quantity can be averaged across preregistration and the RO-Crate format. The appropriate genre is therefore a scoping review with conceptual/framework synthesis, whose output is a map of approaches and an organizing framework - layers L1-L6 and the proposed SDD framework - rather than a pooled effect estimator. The same discipline of claims that this article postulates for empirical research demands that a narrative synthesis not be presented in the statistical guise of a meta-analysis.

3.2. Source databases and their substitution

Owing to the computational environment of the review - a research agent operating in a network sandbox - the three main commercial databases, Scopus (Elsevier), Web of Science (Clarivate), and EconLit (EBSCO), were unavailable. In their place, OpenAlex was adopted as the search foundation: an open index covering more than 250 million works and, in practice, a superset of Scopus and Web of Science coverage, supplemented by arXiv (preprints) and Crossref (publication metadata). PubMed/PMC was likewise deemed unavailable, though its biomedical content reaches the corpus indirectly through OpenAlex. This substitution is a deliberate methodological limitation, discussed further in Section 3.7 and documented in Appendix B (search log); full texts of closed-access works, mainly from Elsevier, were obtained through a manual author channel.

3.3. Search strategy

Searching was conducted in two complementary modes. The first mode, based on topical queries, comprised seven phrase families corresponding to the six control approaches (layers L1-L6) plus the “researcher degrees of freedom” thread in economics, each filtered from 1998 onward with no upper bound. The second mode, a targeted anchor import, introduced fourteen seed works - eleven DOIs and three arXiv identifiers - representing canonical entries for each approach, subsequently expanded through the citation graph, covering both backward references and forward citations. The anchor import deliberately bypassed the lower year filter, so that foundational works predating 1998, the oldest of which dates to 1955, entered the corpus through citation links. The full query chains together with hit counts are documented in the search log in the appendix.

3.4. Inclusion and exclusion criteria

The corpus included works concerning method, credibility, reproducibility, and transparency of empirical research, or the role of AI in research, regardless of discipline. Excluded were works dealing solely with the substantive content of their own field with no bearing on method, as well as off-topic works identifiable by unambiguous title markers, such as PLS-SEM structural equation modeling, biological databases, text-classification datasets, Metaverse-related topics, or purely technical model reports. Exclusions were topical only - never based on language or year of publication. Work types included in the corpus comprised peer-reviewed articles, reviews, preprints, and book chapters, while institutional policies - for example, data and code availability rules of economics journals - were treated as institutional sources cited in the text rather than as corpus items in their own right.

3.5. Screening and deduplication

Screening proceeded in two stages. The first stage, title screening, applied a deterministic dictionary rule: a work was included if its title contained a phrase from one of the layer-vocabulary families, under the overriding rule that a work linked to at least two anchors was always included. The choice of a deterministic rule over classification by a language model was deliberate - a dictionary rule is fully reproducible, recordable in an appendix, and free of run-to-run variability, which stays consistent with the article's own thesis on the reproducibility of research decisions. The second stage, abstract screening, covered works whose title did not contain a key phrase (Tier C layer); for these, abstracts were retrieved from OpenAlex and the same family of dictionary rules was applied, extended with vocabulary that surfaces only in the abstract. Works with no topical signal after both stages were set aside as screened out, though not removed from the register. The full quantitative flow is shown in Figure 1.

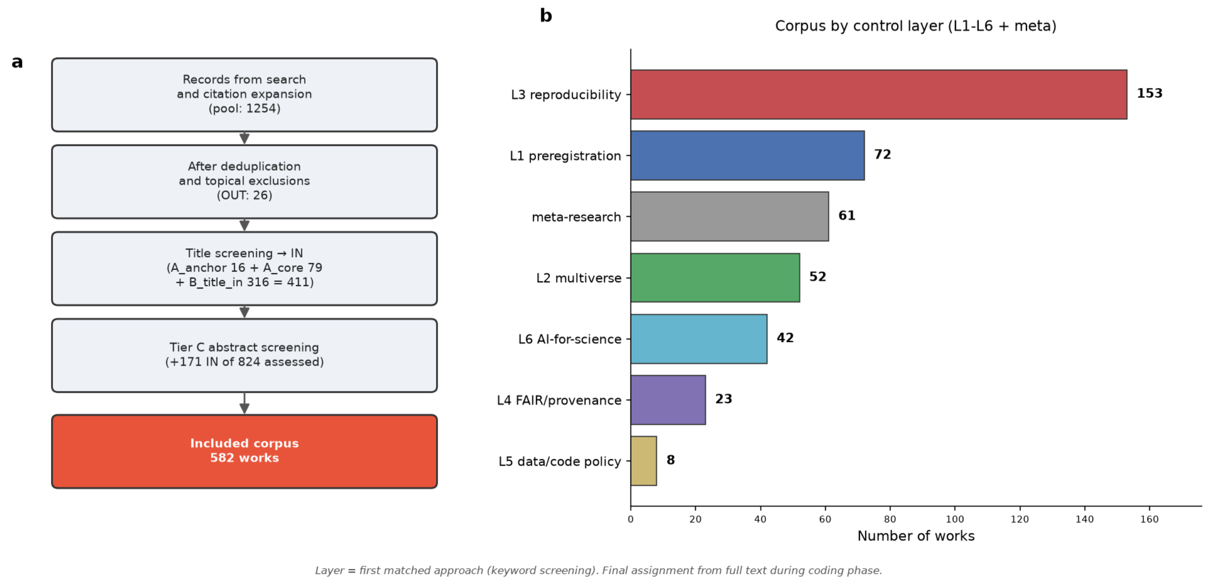

Figure 1. Corpus construction - SDD methodology scoping review (v3, after abstract screening).

Source: own elaboration.

3.6. Coding scheme

Works qualifying as the corpus core were coded using a sixteen-field scheme, covering, among other things, the type of credibility problem, the control mechanism, the point of control - before data collection, before analysis, during analysis, during writing, at submission - the unit of freezing (hypothesis, analysis plan, data, code, artifact, manuscript, prompt), the human role and the AI role, the provenance of data, code, and AI, the approach to exploration, confirmation, and deviations from the plan, transferability to economics and finance, and the potential import of a given element into the proposed framework. The full field specification is given in the coding scheme in the appendix. Coding of interpretive fields relies on full texts and is subject to human adjudication, whereas fields derived from metadata - identifier, source type, year - were populated automatically.

3.7. Method limitations

The method has five main limitations that bear on the interpretation of the results:

1. Substituting OpenAlex for the commercial databases may have missed works indexed exclusively in Scopus, Web of Science, or EconLit, although OpenAlex's practical coverage is broad.
2. Dictionary-based screening, chosen for reproducibility, by definition fails to catch works that are topically relevant but described with different vocabulary; some such works remain in the screened-out layer and do not enter the core.
3. Layer assignment at the screening stage is provisional and sensitive to the order in which phrase families are checked - the final assignment follows only from the full text at the coding stage.
4. The economics data/code policy layer (L5) is the thinnest in the corpus, reflecting both the real distribution of the literature and the limits of dictionary-based search; economic coverage is additionally reported through a measure independent of the layer label, namely the count of works from economics journals.

5. The review does not provide a quantitative synthesis of the strength of evidence, as a meta-analysis would, since the heterogeneity of the corpus precludes this; the conclusions are mapping and organizing in character, and their status is correspondingly qualified in Chapter 8.

All five limitations are explicitly declared, in keeping with the principle of transparency that the article itself addresses. The review so conducted provides the empirical basis for the map of approaches and layers L1-L7, whose characterization is the subject of Chapter 4.

3.8. Declaration of AI tool use

This review was produced with the involvement of AI tools, whose scope we declare explicitly - in keeping both with the journal's policy on AI-generated content and with the article's own postulate of accountability for AI involvement (layer L6). A research agent, Claude Science (built on the Claude Opus 4.8 model), was the primary executor of the technical layer: building and screening the corpus, scaffolding and filling the coding sheet with a quotation and a confidence level for every field, narrative editing of the chapters in both languages, and constructing objects and submission packages. The second, independent coder in the coding-consistency validation was a model from the GPT family (GPT 5.6 Terra), run without access to the primary codes. A separate reasoning model (Claude Opus 4.8) served auxiliary roles - a semantic judge assessing the equivalence of free-text descriptions and a coder of the economic subsample.

All substantive decisions remained with the author as explicit approval gates: freezing the research questions and scope, the design decisions, adjudicating disagreements between coders, accepting the content of each chapter, and the decision to submit. The AI had no access to external accounts and no right to pass a gate without approval. Claims were verified against sources - every quotation in the coding was checked programmatically against the text, and low agreement between the two models was treated as a signal directing the author's attention, not as proof of correctness. The AI tools listed are not, and cannot be, authors of this work; the author bears full responsibility for its content. The versions of tools and models, together with roles and gates, are compiled in the provenance log in the replication package.

4. Six Approaches of Credibility Control

4.0 Introduction: six answers to one problem

The problem sketched in earlier chapters - that the path from data to result conceals hundreds of analytical decisions, of which typically only the combination that "worked" gets reported - is not new. Science has developed a series of responses to it, but each arose in a different discipline, in a different decade, and under a different name. This chapter maps six such approaches as layers of credibility control (L1-L6), describing for each one what it freezes - which decision it renders irreversible or explicit - at which point in the research cycle it intervenes, and what it deliberately does not cover. This last element is central to the article's argument: no single approach covers the entire process, and the boundaries between them mark out the gap that the SDD framework is meant to fill in Chapter 6.

The layering of the works was assigned on the basis of the full text, not the title alone - the coding of 28 works in this cross-cutting map across 16 interpretive fields, described in Chapter 3, is compiled in Table 2 (a separate subsample of 13 economic works, coded with the same scheme, is discussed in Chapter 7).

Table 2. Six approaches of credibility control: what each freezes, when, and what it omits.

Layer	Approach	What it freezes	When	What it omits	Works (corpus)	n
L1	Preregistration / PAP / registered reports	Analysis plan: hypotheses, variable definitions, regression specifications	Before analysis (plan archived before data unblinding)	Computational reproducibility of the pipeline; artefact provenance/ interoperability; accountability of AI involvement	Olken 2015; Ofosu&Posner 2020 (Do PAP Hamper Publication); Hagger i in. 2016; McKiernan i in. 2016	4
L2	Multiverse / specification curve / MHT	Family of hypotheses and testing procedure (multiplicity control)	During analysis (correction at inference)	Pre-specification timing; artefact packaging/ sharing; institutional incentives	Benjamini&Yekutieli 2001; List/Shaikh/Xu 2016	2
L3	Reproducibility / research compendia	Artefact/code/ computational environment for re-execution	During and after analysis (result re-execution)	Whether original decisions were pre-specified (a p-hacked result reproduces faithfully); sharing incentives	OSC 2015; Munafo i in. 2017; Singularity 2017; ARIA 2025	4
L4	FAIR / provenance / RO-Crate	Data + metadata + workflow as findable, reusable artefact	At publication/ dissemination (artefact packaging)	Whether analysis was pre-specified; multiplicity control; real incentive to share	Wilkinson i in. 2016 (FAIR); Bechhofer 2010 (RO); RO-Crate 2022; FAIR workflows 2025; CARE 2020	5
L5	Data/code policies / replication packages / incentives	Manuscript/ data deposit at submission; researcher-assessment criteria	At submission and in downstream evaluation (journal/ institution enforcement)	Internal analytic discipline (freezing decisions); technical pipeline reproducibility	Hong Kong 2020; Wallis 2013; Open Innovation 2020; CLDF 2018; Christensen/Miguel 2019; Giofre i in. 2017	6
L6	AI-for-science / LLM agents / spec-driven AI	Prompt / tool-space / template - frozen agent action-space	During analysis (execution loop) + human checkpoints/ review	Multiplicity discipline; hypothesis pre-registration; data governance - unless explicitly composed in	Coscientist 2023; ChemCrow 2024; Agent Laboratory 2025; AI Scientist 2024; LLM-chem review 2024; LLM-ABM 2024	6
L7	Publication incentives / result-blind review (SDD contribution)	Acceptance criterion based on question and plan quality,	Before results are collected (Stage-1 review) - the	Not a separate corpus approach - the layer SDD adds on top of L1-L6	(pochodna registered reports L1 + polityk L5; nie osobno kodowana)	0

Layer	Approach	What it freezes	When	What it omits	Works (corpus)	n
		not on the result	enveloping layer			

Source: own elaboration.

4.1 L1 - Preregistration, analysis plans, registered reports

Control at the temporal source freezes hypotheses and the analysis plan before the researcher sees the results. The canonical mechanism is the pre-analysis plan (PAP) - a prespecification of variables, data-cleaning rules, and regression specifications, archived with a timestamp prior to the analysis of unblinded data. Olken (2015) provides a full cost-benefit analysis of this kind of freezing in economics, explicitly distinguishing a confirmatory component - prespecified, held to FDA-clinical-trial-level rigor - from an exploratory one, openly labeled and subject to lower rigor [10.1257/jep.29.3.61]. Ofosu and Posner (2020), for their part, show that the publication cost of using a PAP is tolerable, which undercuts the principal argument against its adoption [10.1257/pandp.20201079].

Freezing of the plan can also be enforced collectively. The Registered Replication Report by Hagger et al. (2016) distributes one standardized protocol across 23 laboratories, where preregistration functions as a contract between sites - a strong form of L1 adapted to multi-lab replication [10.1177/1745691616652873].

This approach does not, however, cover the computational reproducibility of the pipeline itself - a plan can be honored while the code remains unreproducible - nor the provenance and interoperability of artifacts, nor accountability for AI involvement. L1 disciplines the researcher's intent, not the execution of the analysis.

4.2 L2 - Multiple testing, multiverse analysis, specification curve

Even under an honest plan, some decisions remain hidden: the space of possible analyses. When the same data can yield hundreds of legitimate specifications, the sheer number of tests inflates the false discovery rate on its own. The answer is formal control of multiplicity. Benjamini and Yekutieli (2001) supply the statistical foundation in the form of a procedure controlling the False Discovery Rate, with a proof of validity under PRDS dependence that makes it applicable beyond the idealized case of test independence [10.1214/aos/1013699998]. List, Shaikh, and Xu (2016) carry this thinking directly into experimental economics, providing familywise-error-rate correction procedures for three multiple-testing scenarios [NBER w21875, 10.3386/w21875].

L2 does not, however, reach back to the moment of prespecification - the correction operates at the inferential stage and does not secure the timing of decisions - nor does it address the packaging and sharing of artifacts or the incentives to apply the control in the first place. This layer disciplines the space of tests, not their sequence or permanence.

4.3 L3 - Reproducibility and research compendia

Freezing the artifact, the code, and the computational environment lets the result be reconstructed. The empirical flashpoint was the Open Science Collaboration (2015) study: a direct replication of 100 psychology studies showed that replication effects were, on average, half the size of the originals, with the entire process built on a preregistered protocol and an independent audit of analyses in R - establishing a template for an auditable replication pipeline [10.1126/

science.aac4716]. Munafò et al. (2017) organize the response into a programmatic reform map spanning five categories - methods, reporting, reproducibility, evaluation, and incentives - a catalogue of mechanisms that the SDD framework is ultimately meant to operationalize [10.1038/s41562-016-0021]. The technical layer is closed out by Singularity (Kurtzer et al. 2017): a container encapsulating the entire computational environment so that the same computation executes identically on a different machine [10.1371/journal.pone.0177459].

This layer also contains the work closest to the article's own thesis - ARIA (2025), a spec-driven, human-in-the-loop framework in which the researcher defines a specification in natural language and AI generates and validates the code. ARIA is inherently a composite work: it was coded under L3 as a reproducible spec→execution pipeline, but it simultaneously realizes L6 as accountable AI involvement with a human in the loop - it thus constitutes, in itself, a miniature of the layer composition that the SDD framework proposes, and it returns as a direct architectural template in Chapter 6 [arXiv:2510.11143].

The boundary of this layer, however, matters critically for the whole map: the reproducibility mechanism itself - the container, the compendium - does not capture whether the original analytical decisions were prespecified, and **a result obtained through p-hacking reproduces just as faithfully as an honest one**. Computational reproducibility is not the same thing as impartiality. It is worth noting a caveat that follows directly from the data: two of the four works in this layer - OSC (2015) and Munafò et al. (2017) - also freeze the analysis plan, thereby combining L3 with L1. This does not weaken the boundary of the mechanism; rather, it shows that the strongest works close it by adding an L1 layer - a thread developed further in Section 5.2.

4.4 L4 - FAIR, research objects, provenance

Data and workflow are frozen as a findable, described, and reusable artifact. An early formulation is the Research Objects of Bechhofer et al. (2010): a semantically rich aggregation of data, methods, and people as a unit of knowledge, in place of the linear, human-readable-only article [10.1038/npre.2010.4626.1]. The FAIR principles (Wilkinson et al. 2016) codify this into four dimensions - Findable, Accessible, Interoperable, Reusable - deliberately extended from data to algorithms and workflows, under the framing of the "machine as a stakeholder" [10.1038/sdata.2016.18]. A mature implementation is RO-Crate (Soiland-Reyes et al. 2022): packaging of artifacts with metadata in Schema.org/JSON-LD, a candidate container for the research package [10.3233/ds-210053]. Wilkinson et al. (2025) close out the layer by applying FAIR directly to computational workflows - that is, treating the analytical pipeline itself as a FAIR object - which touches the very core of SDD [10.1038/s41597-025-04451-9]. The CARE principles (Carroll et al. 2020), meanwhile, add a governance dimension - the question of who holds authority over data control and how the resulting benefits are shared - which purely technical FAIR does not cover [10.5334/dsj-2020-043].

L4 does not, however, settle whether the analysis was prespecified, does not control multiplicity, and does not create any real incentive to share the artifact in the first place. It makes the artifact durable and described, but not thereby impartial or actually shared.

4.5 L5 - Data/code policies, replication packages, incentives

Control shifts from the individual researcher to enforcing institutions - journals, repositories, evaluation systems. Wallis, Rolando, and Borgman (2013) show the realities of science's "long tail": researchers are willing to share data but are rarely asked to, repositories and incentives are lacking, and sharing alone is not enough without enforcement [10.1371/journal.pone.0067332].

That enforcement works is demonstrated by two further studies. Giofrè et al. (2017) measure how journal guidelines - the new statistics, open-science badges - concretely increased the availability of data and materials [10.1371/journal.pone.0175583], while Christensen, Dafoe, Miguel et al. (2019) use data policies as a natural experiment, demonstrating a measurable effect of data sharing on citations - an econometric argument for L4/L5 mechanisms [10.1371/journal.pone.0225883]. On the incentive side, the Hong Kong Principles (Moher et al. 2020) demand that researcher-evaluation systems reward integrity - without which control mechanisms simply go unused [10.1371/journal.pbio.3000737]. CLDF (Forkel et al. 2018) provides a concrete implementation of a format standard with a validator and an ontology [10.1038/sdata.2018.205], and the Open Innovation in Science framework (Beck et al. 2020) situates all of this within the organization of science as a whole [10.1080/13662716.2020.1792274].

This layer, however, does not reach internal analytical discipline - policy compels deposit, not impartiality of analysis - nor does it address the technical reproducibility of the pipeline. L5 disciplines sharing and incentives, not the content of the analysis itself.

4.6 L6 - AI-for-science, LLM agents, spec-driven AI

The youngest layer is the only one to address a risk that the preceding five did not anticipate: AI involvement as a component of method requiring provenance. Coscientist (Boiko et al. 2023) defines a modular prompting system with an explicit action space (GOOGLE/PYTHON/DOCUMENTATION/EXPERIMENT) and decision logging, treating the prompt as a frozen unit [10.1038/s41586-023-06792-0]. ChemCrow (Bran et al. 2024) adds an LLM agent with 18 expert tools, an explicit Thought/Action/Observation reasoning chain, and safety tooling, establishing a template for an accountable agent [10.1038/s42256-024-00832-8]. Agent Laboratory (Schmidgall et al. 2025) embeds human checkpoints at every stage, showing that human involvement substantially improves quality [arXiv:2501.04227]. The extreme case remains The AI Scientist (Lu et al. 2024) - full autonomy with an automated reviewer - whose openly reported risks, such as the hallucination of entire results or the import of unknown libraries, themselves justify the need for freezing and audit [arXiv:2408.06292]. Two reviews, Ramos, Collison, and White (2024) [10.1039/d4sc03921a] and Gao et al. (2024) [10.1057/s41599-024-03611-3], map the space of agents and identify human alignment and evaluation as open challenges of accountability.

L6 does not cover multiplicity discipline, hypothesis preregistration, or data governance unless these are explicitly attached to it. It disciplines AI involvement, but does not itself inherit the safeguards of layers L1-L5.

4.7 A cross-cutting layer: diagnosis of the problem

Beyond the six layers, the corpus contains one work that is not a layer of control but a diagnosis of the problem all six are trying to solve. Ioannidis (2005) shows, through a probabilistic model, that the positive predictive value of a result depends on test power, the significance level, and the pre-study odds - meaning that a single $p<0.05$ from one study is insufficient [10.1371/journal.pmed.0020124]. This is the motivation for the entire map proposed in this chapter: if a single test were sufficient, none of the layers L1-L6 would be needed at all.

The six approaches described above arose independently of one another, each addressing a different moment of the research process in turn - from the preregistration of hypotheses, through control of the test space and computational reproducibility, to data governance and AI accountability. Chapter 5 shows how these layers co-occur in research practice, and where, in this mosaic, gaps remain that no single approach closes.

5. Comparative synthesis

Chapter 4 presented six approaches of epistemic control separately, each in its own historical and methodological logic. This chapter sets them side by side in a single comparative matrix, in order to answer the two questions posed in the introduction: what control mechanism do they share (RQ2), and what do they, taken together, fail to cover (RQ3)? The article's thesis is that the gap lies not within any one layer, but between them - and that this can be shown, not merely asserted.

5.1 The shared mechanism: freezing a decision as an auditable object

Set alongside one another, the six approaches reveal a single pattern. Each renders some decision irreversible, or explicit, at a specific point in the research cycle, so that it cannot quietly be altered to fit the outcome: L1 freezes intention - hypotheses, the study plan - before data collection; L2 freezes the test space, meaning the family of hypotheses and the correction procedure, at the point of inference; L3 freezes execution, that is, code and computational environment, so that it can be reproduced; L4 freezes the artifact - data and workflow - as a described, persistent object; L5 freezes release, meaning deposit and evaluation criteria, through institutional enforcement; and L6 freezes the involvement of AI, understood as prompt and tool space, as a logged unit. The common denominator of these six mechanisms is the freezing of a decision into an object that can later be audited; they differ only in which decision each freezes, and at which point of the research cycle. This pattern becomes visible only in the approach × control-dimension cross-tabulation - what each layer freezes, when, how it is audited, and what it fails to cover - which Table 3 sets out.

Table 3. Synthesis matrix: six approaches by control dimension (what it freezes / when / how it audits / what it omits).

Layer	What it freezes	When	How it audits	What it does NOT cover
L1 Preregistration / PAP	Intent: hypotheses, analysis plan	Before data	Archived plan vs report	reproducibility, artefacts, AI
L2 Multiverse / MHT	Test space: hypothesis family	At inference	Corrected p / specification curve	prespec. timing, packaging, incentives
L3 Reproducibility / compendia	Execution: code, environment	During and after analysis	Code re-execution	whether decisions prespec. (p-hacking reproduces)
L4 FAIR / provenance	Artefact: data+metadata+workflow	At dissemination	Provenance metadata, workflow graph	prespecification, multiplicity, incentives
L5 Data/code policies / incentives	Sharing: deposit, assessment criteria	At submission and downstream	Institutional enforcement (data editor)	internal analytic discipline, reproducibility
L6 AI-for-science / LLM agents	AI involvement: prompt, tool-space	In execution loop + checkpoints	Log of prompts/ agent actions	multiplicity, preregistration, governance

Source: own elaboration.

5.2 The layers are not disjoint - and that is a clue

Coding the corpus revealed something important for the article's thesis: some works belong to more than one layer at once. Two of the four works classified as L3 (Open Science Collaboration 2015; Munafò et al. 2017) freeze not only code but also the analysis plan, thereby joining L1 to L3. FAIR (Wilkinson et al. 2016) extends from data to workflow, already touching L3. CARE (Carroll et al. 2020) adds to L4 a governance dimension proper to L5. Coscientist and ChemCrow, classified as L6, in turn build in provenance logging that in spirit belongs to L4.

This is not coding noise; it is a signal. The best work already composes layers, because no single layer suffices on its own - but the composition happens ad hoc, within the confines of a single paper or a single tool, without any shared architecture. The SDD thesis, developed in Chapter 6, generalizes this observation: since the layers must be combined in any case, they should be assembled deliberately into a single workflow, rather than left to chance.

5.3 Four gaps visible from the matrix

The "what it does not cover" column of the matrix, read down its length, exposes four decisions that none of the six layers freezes systematically. The first is AI provenance: only L6 addresses the involvement of artificial intelligence at all, and it does so as a fresh, immature thread, since the five older layers took shape before prompts, models, and agents became part of method - a prompt as a logged, versioned unit simply has no place in L1-L5. The second is claim discipline: no layer ties the language of conclusions to the status of the evidence behind them, and while Ioannidis (2005) diagnoses the consequence of this state of affairs - the overinterpretation of a single $p<0.05$ - none of the approaches freezes the rule that a claim cannot be stronger than the artifact underpinning it. The third is the construction of the text from artifacts: L3-L4 make artifacts reproducible and described, but the manuscript still comes into being alongside the results, as a separate text transcribed by hand, and no one freezes the dependency whereby a number in a sentence corresponds to a number in an artifact. The fourth, finally, is human approval gates: L6, in the example of Agent Laboratory, shows human checkpoints as a single instance, but there is no approach that makes an explicit human/AI decision gate a standing element of the architecture.

These four gaps are not the author's opinion - they follow directly from the empty cells of the matrix, and they constitute the specification of what an SDD framework must add, which is the subject of Chapter 6. Reading the empty cells as gaps assumes at the same time that the corpus is representative of each approach - a limitation of dictionary-based selection, discussed in §3.7 and Chapter 8. The map of seven layers of epistemic control, shown in Figure 2, presents these gaps as the territory that only a seventh layer - publication incentives and result-blind review - together with the SDD architecture, finally closes.

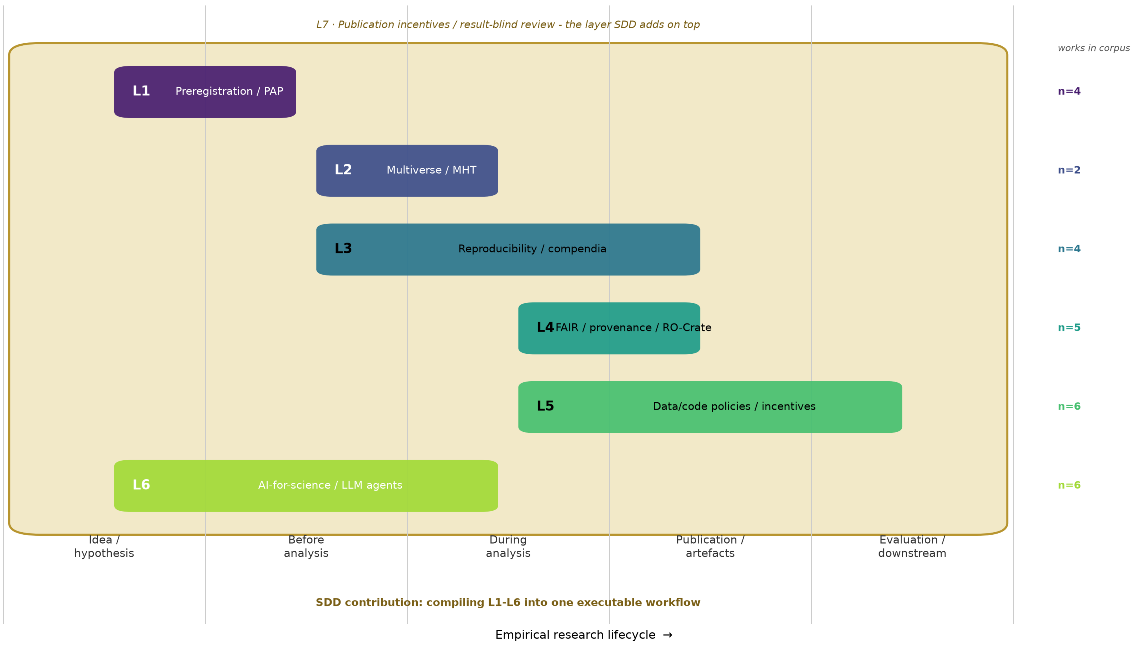

Figure 2. Map of seven epistemic-control layers: where in the research lifecycle each approach freezes decisions.

Source: own elaboration.

5.4 A response to an anticipated objection

The strongest objection that can be raised against this article is that it amounts to nothing more than preregistration plus reproducibility under a new name. The matrix answers it. Preregistration (L1) freezes neither code nor AI involvement; reproducibility (L3) freezes neither intention nor the language of claims; together, the two approaches cover none of the four gaps discussed in §5.3. What SDD adds, then, is not a relabeling of existing practices but the assembly of six layers into a single workflow, together with the four missing freezes. The elaboration of this originality - and a detailed account of the architecture that realizes it - is the substance of Chapter 6.

6. The SDD Framework (Proposed Methodology)

Chapter thesis

Chapter 5 showed that the six approaches share a single mechanism - freezing a decision as an auditable object - and leave four gaps: AI provenance, claim discipline, artifact-driven manuscript construction, and human gates. This chapter presents spec-driven development (SDD) for empirical research as an **architecture of epistemic control** that composes six layers into one executable workflow and adds the missing quartet of freezes as a seventh layer. The distinction that matters, and that must be defended against a reviewer, is this: SDD is **not** a new single-layer technique - it is a compiler that makes executable what the approaches already achieve in scattered, ad hoc form.

Seven layers of epistemic control

SDD organizes control into seven layers. The first six are the approaches discussed in Chapter 4, translated from separate practices into coordinates of a single architecture; the seventh is the framework's own original contribution.

- **L1 intent** - the spec freezes hypotheses and the analysis plan before the data are seen (inherited from preregistration/PAP).
- **L2 analysis space** - the spec declares the family of tests and the multiplicity-correction rule (inherited from the multiverse/MHT approach).
- **L3 execution** - the pipeline and environment are frozen as reproducible (inherited from reproducible research).
- **L4 artifact** - data, code, and workflow are packaged with provenance and a persistent ID (inherited from FAIR/RO-Crate).
- **L5 disclosure** - the submission package complies with journal policies and their enforcement (inherited from data/code policies).
- **L6 AI involvement** - prompts, models, and agents are logged and versioned as method components (inherited from AI-for-science).
- **L7 incentives** - the acceptance criterion is based on the quality of the question and the plan, not on the outcome (result-blind / registered reports, functioning as the layer that binds the rest together).

Beyond this inheritance, SDD adds four freezes that no layer possessed on its own (cf. §5.3): AI provenance as a mandatory artifact rather than an option; claim discipline, in which the language of a conclusion is bound to the status of the evidence behind it; deterministic manuscript construction from artifacts, where every number in the text corresponds to a number in an artifact; and explicit human/AI approval gates between stages. Table 4 sets each layer against what SDD adds on top of the L1-L6 approaches.

Table 4. Seven epistemic-control layers in the SDD framework and what SDD adds beyond each approach.

Layer	Dimension	Approach	What SDD adds
L1	Intent	Preregistration / PAP	spec as executable input artefact
L2	Analysis space	Multiverse / MHT	test-family declaration in the spec
L3	Execution	Reproducible research	pipeline bound to spec and AI log
L4	Artefact	FAIR / RO-Crate	artefact = source of every claim
L5	Dissemination	Data/code policies	submission package generated from inventory
L6	AI involvement	AI-for-science	AI provenance MANDATORY, not optional
L7	Incentives	result-blind / reg. reports	acceptance criterion = plan quality

Source: own elaboration.

Minimal workflow: seven stages with gates

The layered architecture becomes a method only once it is translated into an ordered sequence of stages with approval gates. The minimal SDD workflow comprises seven stages:

1. **Idea** - the research question and thesis stated in natural language (the input spec).
2. **Design freeze** - hypotheses, analysis plan, and test space are frozen (L1+L2); a human gate precedes access to the data.
3. **Data freeze** - a data snapshot is frozen together with source provenance and variable definitions (L4); for secondary data, this includes vendor version, retrieval date, and revision rules.
4. **Analysis artifacts** - the pipeline runs inside the frozen environment; every result is an artifact with a log, including a log of AI prompts and models (L3+L6).
5. **Inventory** - artifacts are inventoried: what was produced, from what, with which tool; a claim-discipline gate requires every claim to point to an artifact.
6. **Manuscript build** - the manuscript is assembled deterministically from artifacts, with numbers, tables, and figures inserted from the inventory rather than transcribed by hand.
7. **Submission package** - a package compliant with the target policy (L5), accompanied by full provenance; a human gate precedes submission.

Explicit human gates stand between stages 2/3, 3/4, and 5/6 - points at which the researcher approves the transition and which AI cannot bypass: before data access, once the plan is frozen; before the pipeline runs, once the data are frozen; and before the manuscript is built, once the artifacts have been inventoried. This operationalizes the human-in-the-loop principle that, at layer L6 - in Agent Laboratory or ARIA - appeared as a single instance; here it functions as a permanent architectural feature, as illustrated in Figure 3.

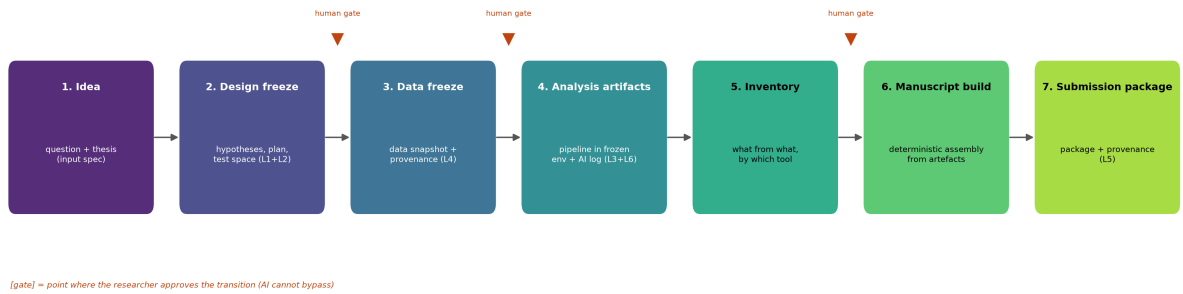

Figure 3. Minimal SDD workflow: seven stages with human approval gates.

Source: own elaboration.

Light and full variants

SDD does not assume that every study needs the full machinery. The light variant is limited to a project spec (L1+L2), a data snapshot (L4), a reproducible script (L3), and a list of claims pointing to artifacts, which implements claim discipline - without containers and without AI agents; it suits a single researcher, public data, and a simple analysis. The full variant covers all seven layers, including an environment container, AI provenance logging, deterministic text construction, and a submission package - and it is this variant that answers the needs of teams, sensitive or licensed data, and pipelines involving LLM agents.

Both variants nonetheless share a common, non-removable core of three freezes: the plan before the data, the artifact as the source of the claim, and provenance of what the AI did. The rest of the architecture then scales with the risk profile of the specific project.

Originality explicitly distinguished from L1-L6

This is the gate for the author's contribution. The originality of SDD does not lie in inventing a new layer, but in five elements, each absent when the approaches are considered separately:

1. **Workflow-level integration.** SDD shows that L1-L6 are not competing schools but coordinates of a single architecture; the composition that occurs ad hoc across the corpus (§5.2) becomes an explicit, executable sequence of stages.
2. **AI provenance as an obligation, not an option.** The prompt, the model, and the agent are components of the method requiring a log on par with the code - something the five older layers did not provide for.
3. **Manuscript construction from artifacts.** The manuscript is a deterministic product of the inventory rather than text written alongside the results, closing the gap between the reproducible artifact and the manually transcribed number.
4. **Claim discipline.** The language of conclusions is explicitly bounded by the evidentiary status behind the claim.
5. **Human gates as architecture.** Human/AI approval is a fixed feature between stages, not good practice left to the researcher's discipline.

None of these five elements is systematically frozen by L1-L6, as confirmed by the empty cells of the matrix in §5.3. At the same time, none of them is "novelty out of nothing" - each constitutes the logical closure of a direction in which the approaches were already heading. It is in precisely this sense that SDD is a synthesis at the workflow level, rather than a separate invention.

The distinction between a compiler and a single-layer technique is, moreover, strict rather than rhetorical. A layer technique - such as a specification curve in L2 or a container in L3 - solves one control problem at one point in the process and is interchangeable with another technique of the same layer without disturbing the rest. The compiler does not add a sixth technique alongside them; it fixes the ordering, dependencies, and gates between layers so that the output of one becomes the frozen input of the next: plan before data, data before analysis, artifact before claim. The contribution lies in the composition, not in the components - the same techniques run without an enforced ordering and gates yield the ad hoc practice that the corpus shows to be the rule, not the exception. Separately, layer L7 should be distinguished from registered reports, from which it derives: registered reports are a journal institution, in which the editors commit to evaluating and accepting a study on the basis of its plan before results are gathered, whereas L7 in the SDD framework is a layer of the study's architecture that the researcher can maintain regardless of whether the target journal offers such a track. SDD does not replace registered reports where they are available - it operationalises their principle as a layer that works even without them.

Limitations of the framework

Honesty toward the consistency test requires naming the boundaries of this proposal. First, SDD is a framework, not an evaluation - the present article is a scoping review and does not provide empirical evidence that applying SDD raises reproducibility; that is a task for a separate study. Second, there is a cost: the full variant adds overhead that, for simple studies, may exceed the benefit, hence the need for the light variant. Third, AI provenance depends on tools that change rapidly - the framework specifies *what* to log, not *with what*, and this abstraction may date over time.

Adapting the framework to economics and finance - to secondary, market, and macroeconomic data - is the subject of Chapter 7.

7. Application in economics and finance

7.1 Why economics and finance make a good testing ground

Chapters 4-6 built the SDD framework on cross-cutting literature, layer layer by layer and approach by approach. This chapter carries it into research practice in economics and finance, addressing the question of why the framework gains rather than loses precision on this ground. Economics is a particularly favorable field for three reasons. It already has a mature culture of preregistration and data policies, so adaptation does not start from zero. Its data are also mostly secondary - market, macroeconomic, panel data obtained from providers - which shifts the burden of control from the question "was the data collected honestly?" to the question "was it processed honestly?", and this is exactly the point at which SDD operates. Finally, the corpus underlying this review includes 71 works from economics venues out of 582 qualified items - a coverage metric derived from phase 01-02 - among which 19 coded works come from economics venues (out of 41 coded in full text, a purposive selection by layer) and serve as empirical anchors for the whole chapter.

7.2 Preregistration and registries in economics as the upper gate

Economics already has its own well-developed preregistration practice, which SDD inherits as layer L1 rather than introducing from scratch. Olken (2015) provides the canonical analysis of this mechanism: the pre-analysis plan, transferred from clinical trials into economics, with an explicit separation of the confirmatory - prespecified - part from the exploratory part, which corresponds directly to the architecture of the SDD upper gate, namely design freeze [10.1257/jep.29.3.61]. The main resistance to adopting such plans has always been the presumed publication cost, yet Ofosu and Posner (2020), analyzing a sample of top-5 journal practice, show that this cost is tolerable, thereby removing a key argument against preregistration [10.1257/pandp.20201079]. For an economist, an analysis plan deposited in a registry - the AEA RCT Registry, OSF - is already a ready-made implementation of L1; SDD adds only one requirement to it: that this plan be an executable input artifact, not a document existing alongside the actual analysis. The economic subsample coded in this phase adds three works directly on pre-analysis plans. Banerjee, Duflo, Finkelstein, Katz, Olken, and Sautmann (2020) argue against overly rigid prespecification: the plan should freeze the confirmatory part but not suppress exploration [10.3386/w26993] - which directly supports the SDD architecture, in which the confirmatory part is frozen and the exploratory part explicitly flagged rather than eliminated. Miguel (2021) offers a synthetic review of the adoption of open data, preregistration, and journal policies in economics, with evidence that the worst fears about adoption costs did not materialize [10.1257/jep.35.3.193]. The p-value debate (2023) in turn shows the heterogeneity of inference rules across journals, motivating an explicit declaration of the rule in the specification at the L1 and L7 level [10.30430/gjae.2023.0231].

7.3 Controlling test multiplicity

Experimental economics confronted the problem of researcher-degrees-of-freedom earlier than many other disciplines. List, Shaikh, and Xu (2016) supply familywise-error-rate correction procedures for three multiple-testing scenarios in economic experiments, a direct realization of layer L2 in the language of econometrics [10.3386/w21875]. The statistical foundation of this control - the Benjamini-Yekutieli method for FDR, valid under test dependence - is routinely applied in econometrics when screening multiple factors [10.1214/aos/1013699998]. SDD contributes no new statistics here; it merely requires that the family of tests and the correction rule be declared in the specification before the analysis, rather than chosen only after the results have been seen. The subsample adds two anchors. Romano and Wolf (2005) supply a canonical stepwise FWER-control procedure for financial data - funds, CAPM models, VaR measures - as formalized data snooping, a realization of L2 in the native language of econometrics [10.1111/j.1468-0262.2005.00615.x]. Borusyak, Jaravel, and Spiess (2024) in turn separate the estimand and its assumptions, frozen a priori, from the estimation itself, supplying a working code artifact - a model of layers L2-L4 tied to the upper gate [10.1093/restud/rdae007].

7.4 AEA data/code policies as the lower gate

The most developed element of control infrastructure in economics is the data/code policy of journals and the institution of the data editor, entrenched at the AEA and at Econometrica. This is a mature L5 layer - the lower gate, at which the replication package becomes a condition of publication. That such enforcement genuinely changes research practices is demonstrated by works from the corpus: Giofrè and coauthors (2017) measure the effect of journal guidelines on data availability [10.1371/journal.pone.0175583], and Christensen, Dafoe, Miguel, and others (2019) - economists themselves - use data policies as a natural experiment, showing a measurable effect of data sharing on citations [10.1371/journal.pone.0225883]. Crucial for the framework, however, is the observation that the AEA policy is a downstream gate, positioned at publication, whereas SDD adds upstream gates - before the data, before the analysis. A replication package checks whether a result can be reproduced, but it does not check whether the analytical decisions were prespecified, and this is the same boundary we already encountered with layer L3 in section 4.3: the AEA package will reproduce p-hacking just as faithfully as it reproduces an honestly conducted analysis. Two works from the subsample provide hard evidence of this boundary. Chang and Li (2015) reproduced the key result of only 33 percent of sixty studies without contacting the authors, and even with their help fewer than half [10.17016/feds.2015.083] - evidence that a deposit policy alone does not suffice without enforcement and the standardization of executable artifacts. Duvendack, Palmer-Jones, and Reed (2017) show in turn that economics lags behind psychology and political science in its replication culture [10.1257/aer.p20171031], which strengthens the case for upstream gates, not only downstream ones.

7.5 Upstream workflow for financial and macroeconomic data

This is where the chapter's proper contribution lies. Financial and macroeconomic data carry risks that standard data/code policies do not capture, because these are risks of secondary-data processing that arise before the analysis, not at the moment of publication. SDD addresses them at the data-freeze stage described in section 6.2, and they can be reduced to four concrete mechanisms.

- **Look-ahead bias** - the use of information that was not available at the moment of the decision, for instance data revised retrospectively or indicators published later. The freeze that prevents this is a point-in-time-stamped data snapshot: the date as of which the data were known, not the date on which they were retrieved.
- **Survivorship bias** - a sample restricted to entities that survived (funds, companies), because databases remove expired entities. The freeze here is an explicit declaration of the universe, including entities that dropped out, together with the rules for their inclusion in the sample.
- **Data revisions** - macroeconomic series such as GDP or employment are revised after first publication, so that an analysis run on “final” data differs from one run on “real-time” data. The freeze is the vintage version of the series, taken from a real-time database, rather than its current reading.
- **Provider definitions** - the same nominal indicator differs across providers, for instance Bloomberg, Refinitiv, or CRSP, and also across versions from the same provider. The freeze here consists of recording the provider, the definition version, and the extraction date as part of the data artifact’s provenance.

The empirical foundation for this argument is the work of Menkveld and coauthors (2024): 164 research teams tested the same hypotheses on the same data, and the dispersion of their results - nonstandard errors - proved larger than the standard error of estimation [10.1111/jofi.13337]. This is measurable evidence that researcher degrees of freedom generate uncertainty that freezing the analysis path and a two-stage peer feedback reduce - the strongest empirical anchor for the whole chapter. The common denominator of the four risks below is that each constitutes a silent decision about input data, one that SDD renders an explicit, frozen artifact with documented provenance - a realization of layer L4 (data artifact), tied to the upper gate requiring a human decision before the analysis begins. Table 5 maps the seven SDD layers onto successive stages of a study conducted on financial data, indicating the specific freeze and the risk it addresses.

Table 5. Mapping of SDD layers onto the workflow stages of empirical research in finance.

Layer	Stage	Mechanism	Freezes	Risk
L1	Design freeze	Registered PAP (AEA/OSF)	hypotheses + analysis plan	HARKing (hypotheses after results)
L2	Design freeze	test-family declaration	hypothesis family + MHT correction	untamed multiplicity
L4	Data freeze	point-in-time data snapshot	vendor + version + extraction date	look-ahead, vendor definitions
L4	Data freeze	universe declaration	delisted entities + rule	survivorship bias
L4	Data freeze	series vintage	real-time database vintage	macro data revisions
L3	Analysis artifacts	container + pipeline	environment + econometric code	non-reproducible computation
L6	Analysis artifacts	AI prompt/model log	prompt + model version	hidden AI involvement (e.g. ABM)
L5	Submission	replication package (AEA)	data/code package	no downstream replication
L7	(cross-cutting)	acceptance criterion	plan quality, not result	publication bias

Source: own elaboration.

7.6 The role of AI in economic research

Economics increasingly draws on LLM agents - for literature review, for generating econometric code, and, in agent-based modeling, directly as a component of the method itself. Gao and coauthors (2024) show LLMs functioning as the core of agent-based simulations of markets and societies - an application in which the prompt and the model constitute the specification of the experiment [10.1057/s41599-024-03611-3]. This makes layer L6, namely AI provenance, not an abstract postulate but a concrete requirement: if a result depends on the prompt and the model version, both must be logged and versioned on a par with the code. Standard AEA policies do not yet require this - a gap that SDD fills ahead of time. The economic anchor here is Charness, Jabarian, and List (2023): LLMs as assistants at the design, implementation, and analysis stages of economic experiments, with the freezing of the researcher-AI interaction - the prompt log as an appendix - as an auditable artifact [10.3386/w31679].

7.7 Limits of the adaptation

Honesty toward one's own findings requires three caveats. The chapter proposes an upstream workflow but does not empirically demonstrate that it reduces bias - that is a task for a separate study conducted on real financial data. Some of the risks, especially those tied to point-in-time data and real-time versions, require a data infrastructure that does not exist everywhere, and for some historical series a point-in-time snapshot is simply unavailable. Finally, the 19 coded works from economics venues are a purposive selection from 71 economics-venue items in the corpus - a sample of open-access works, with 33 items behind paywalls beyond the reach of this phase, a limitation of the full-text acquisition channel consistent with the database substitution described in §3.7 - not a representative review of economic practice. The anchors illustrate the transfer of the framework layer by layer; they do not measure the frequency of practices in the discipline, and the sample is purposive and proportional to layers, not random. The metric of 71 works from economics venues measures the corpus's thematic coverage, not coding depth: full text was coded for 41 works, of which 19 come from economics venues.

This transfer of the framework to a specific discipline raises a more general question: how does SDD hold up against its own methodological limitations, and what boundaries of applicability does it reveal beyond economics alone - this is the subject of the next chapter.

8. Risks and open questions

The framework built in Chapters 6-7 - seven SDD layers, a workflow binding them into a coherent research process, and its translation into economics - is a proposal, not a validated standard. Honesty toward the reader requires laying out its internal tensions before a reviewer does so instead. This chapter answers part of the fourth research question from the side of limits: not "what does the framework add" but "where might the framework fail, or cause harm." Each of the five risks is discussed together with a possible direction of resolution - not in order to close it off, but to show that these are design problems for further work, not objections that invalidate the proposal as a whole.

Does an overly rigid workflow inhibit discovery?

This is the most serious substantive objection that can be raised against the framework. Freezing the analysis plan ahead of the data at the L1 level protects against HARKing, but genuine discovery is often unplanned - the most interesting result is frequently the one nobody anticipated. A workflow that penalized departure from specification would risk killing exploration rather than disciplining it.

The framework addresses this tension not by abolishing the gate but by explicitly separating modes. Here SDD inherits from the multiverse-analysis approach at the L2 level, and from Olken's (2015) practice of separating the confirmatory - that is, prespecified - component from the exploratory one. Exploration remains permitted, and indeed desirable, provided it is labeled as exploration and its results are not presented in the language of confirmation. The rigidity therefore concerns not what may be investigated, but how strongly one may claim it. What remains open is how to design a gate that would permit re-specification - a new, explicitly dated plan - without thereby opening a back door to silent p-hacking; this is a protocol-design problem that the present article does not resolve.

How should exploratory findings be classified?

This is a direct consequence of the preceding tension. Since exploration is permitted, a discipline of conclusion-language becomes necessary - the fifth originality point of the framework: an exploratory finding must carry a label of evidentiary status. The problem is real because the boundary is often blurred - an analysis begun as confirmatory can generate a side question, and the author faces a natural temptation to promote it to the status of a finding.

The solution appears to lie in making the status of a result - confirmatory, exploratory, hypothesis for further testing - a metadata field of the resulting artifact, frozen together with it rather than appended only at the manuscript-drafting stage. This shifts classification from textual rhetoric to data provenance, in keeping with the principle that text is a product of artifacts - the fourth originality point of the framework. What remains open, however, is who adjudicates borderline cases and by what criterion; this remains a human gate, not an automatic rule.

When is an AI verifier sufficiently independent?

Layer L6 postulates independent verification - for example, a second agent checking code or output. The problem is that two models from the same family share the same systematic errors: an “independent” verifier built on the same LLM as the generator may in fact be confirming its own hallucinations. AI Scientist (Lu et al. 2024) explicitly warns that the system can hallucinate entire results; a verifier drawn from the same source is then not a safeguard but an echo.

A direction of resolution is to treat verifier independence as a dimension to be declared, not assumed. AI provenance, the third originality point of the framework, should log not merely that verification took place, but what it consisted of: a different model, a different model family, a human, or a deterministic tool such as code unit tests. Genuine independence increases by degree - from “the same model,” through “a different family,” to “non-AI verification” - and the article does not settle where on this scale the threshold of sufficiency lies; this is an empirical question, dependent on domain and on the stakes of the result.

How should AI involvement be disclosed in the manuscript?

Journal standards for declaring AI use are still in their infancy and remain inconsistent. The framework requires AI provenance - prompts, models, versions - yet no agreed format for disclosing it yet exists; AEA data/code policies at the L5 level do not cover this, as already noted as a gap in Subsection 7.5.

The solution appears to lie in treating AI disclosure as an extension of the replication package rather than as a separate prose section. Since prompts and model versions are already frozen as artifacts at the L6 level, disclosing them amounts to appending a log to the submission package - the same infrastructure that already handles code and data. What remains open, however, is how to reconcile this with model non-determinism, where the same prompt yields different outputs, and with provider-side API version changes: provenance can fix the prompt in place, but it will not always allow the exact output to be reproduced.

How should the effectiveness of the methodology itself be measured?

This is a meta-question, and at the same time the hardest limitation of the entire proposal. The article claims that SDD raises research credibility, but it does not furnish evidence for this - the framework has been tested on a single instance, the making of this very article, and not validated on a sample. This is not false modesty: absent measurement, “effectiveness” remains a design claim, not an empirical one.

The direction of resolution, developed in Chapter 9, runs through empirical validation on several independent research instances - comparing studies conducted with and without the framework in terms of reproducibility, detectability of departures from plan, and the false-positive rate of results. This, however, requires a metric the article does not yet have, and a sample not yet collected. The honest answer is therefore that SDD’s effectiveness is a hypothesis to be tested, not a finding of this review.

Limitations of the review itself

Beyond the tensions within the framework, honesty requires naming the limits of this work as a review. Four seem key:

1. **Search base.** The corpus was built on OpenAlex, arXiv, and Crossref, not on Scopus, Web of Science, or EconLit, which remain behind a paywall. This changes coverage: OpenAlex has broad reach but a different indexing structure than WoS, so some economics work from closed sources may not have entered the analysis. The metric of 71 papers with an economics venue therefore measures coverage within this particular corpus, not across the literature as a whole.
2. **Coding depth.** Forty-one papers were coded full-text (28 in the cross-cutting L1-L6 map from phase 02, shortlisted from 56 and 582 screened, plus 13 in the economic subsample from Chapter 7) - the remainder would have required manual access to content behind a paywall. Claims about the layers therefore rest on a purposive sample, not an exhaustive one.
3. **Status of the methodology.** SDD carries the status `provisional` - it has been tested on a single instance, not validated as a standard, as discussed more fully in the subsection on measuring the effectiveness of the methodology.
4. **AI involvement in producing the text.** The article was produced with the use of an AI assistant (Claude Science): corpus searching, interpretive coding with citation verification, and prose drafting. In keeping with the framework's own logic, at the L6 level this involvement is disclosed rather than concealed - the process is documented in the `_LOG_procesu/` log in accordance with the FSAIM methodology [10.2139/ssrn.6853581] under which this article was produced. Interpretive coding, though programmatically verified against the source text, carries a risk of error, which the human author supervises through adjudication gates.

These four limitations do not invalidate the contribution of this work - the mapping, the synthesis, the transfer of the framework into economics - but they do delimit its scope: this is a proposed framework grounded in a review of defined coverage, not a definitive audit of the field. This is precisely what motivates the direction of the following chapter, in which these open questions - empirical validation foremost among them - become a research program rather than merely a list of caveats.

9. Conclusion

9.1 SDD as a compiler of good practice

The previous chapter closed the discussion of the method's limitations set out in §8.5; what remains here is to draw the threads together into a single answer to the question posed at the outset of this article. The review began from a quiet vulnerability in empirical research: hundreds of small analytic decisions lie on the path to any result, and only the combination that "worked" is ever shown. Science has built separate defences against this - preregistration, multiverse analysis, code and data reproducibility, provenance standards, data/code policies - but each protects only one fragment of the process, and these defences are scattered across disciplines that rarely read one another.

Mapping these approaches onto six layers (Chapter 4, RQ1) and synthesising them (Chapter 5, RQ2) showed that they share a single underlying mechanism: the freezing of a decision as an auditable object at a defined point in the process. This is the axis on which the article's thesis rests. SDD is not another, separate safeguard, nor - contrary to a first impression - a tool for writing or coding with AI; it is an architecture of epistemic control, a way of working that assembles scattered approaches into a single executable workflow with human approval gates between stages (Chapter 6, RQ3).

What the framework adds on top of the synthesised layers (RQ3-RQ4) are four elements that no single approach possessed together: accountability for AI involvement, understood as prompts, models, and agents treated as provenance objects; the construction of the manuscript as a deterministic product of artifacts; explicit human gates between stages; and a discipline of conclusion-language tied to evidentiary status. Transposing the framework to economics and finance (Chapter 7, RQ5) showed, in turn, that it gains the most precisely where data are secondary - because the burden of control then shifts from the question "was the data collected honestly" to the question "was it processed honestly," and this is a domain in which freezing data-related decisions yields a concrete, measurable benefit.

Hence the metaphor that closes the whole: SDD is a compiler of good practice. It does not invent credibility controls from scratch - it compiles existing, well-tested approaches into a single run, adding the one element missing from all of them: AI accountability. The strongest formulation to defend remains the one from the first chapter: *SDD for empirical research is a workflow-level synthesis of open science, reproducible research, provenance standards, and AI accountability - not merely an AI-assisted writing or coding method.*

9.2 A response to the principal objection

The most serious objection that can be raised against this proposal is that it is merely preregistration plus reproducibility under a new name. The review answers it not by declaration but through the substance of Chapters 5-8. Preregistration (L1) does not cover computational reproducibility; reproducibility (L3) does not cover pre-specification; and neither layer on its own covers AI accountability, the construction of text from artifacts, or the discipline of claim formulation. The value of SDD lies exactly in the joint between them - in the fact that it enforces all layers at once and adds those that were absent from every existing arrangement. A compiler, after all, is not "merely the sum" of the libraries it links; it is what makes them executable together.

9.3 Direction: empirical validation

This article is a review and a proposal, not proof of efficacy - a limitation stated explicitly in §8.5. The natural next step remains the empirical validation of SDD across several independent research instances: comparing studies conducted with and without the framework in terms of the reproducibility of results, the detectability of departures from the frozen plan, and the rate of false-positive findings. This requires a metric of methodological effectiveness that this article does not yet propose, and a sample of studies that has not yet been assembled. Until such validation is carried out, the framework's status remains `provisional` - tested on the single instance constituted by the production of this text under the FSAIM methodology [10.2139/ssrn.6853581], but not established as a standard.

To make the framework falsifiable, we state three predictions that a future study may confirm or refute. First, on reproducibility, studies conducted with the full SDD variant should attain a higher rate of independently reproduced results than matched studies without the framework - if the difference is zero or negative, the framework's contribution to reproducibility is refuted. Second, on the detectability of departures from the plan, departures from prespecification in studies with a frozen analysis plan (L1+L2) should be detectable from the artifacts alone, without access to the author's knowledge - if an auditor cannot distinguish prespecified from exploratory analysis better than chance, the plan-freezing layer does not deliver the auditability it promises. Third, on AI accountability, the presence of a mandatory provenance log (L6) should raise the detectability of AI-origin errors such as fabricated citations or model drift - if the detection rate does not rise, the provenance log is a cost without an audit benefit. Each prediction names a measurable quantity and a condition that falsifies the corresponding element of the framework; their operationalisation is the subject of the announced separate validation study, not this review.

This is an honest closing point: the review has mapped the field, synthesised it into a framework, and shown where that framework reaches further than its constituent parts. Whether it reaches effectively is a question that only measurement, not argument, can answer. The article sets out the framework and indicates how it might be refuted or confirmed; the rest belongs to the data.

References

1. Banerjee, Abhijit, Duflo, Esther, Finkelstein, Amy, Katz, Lawrence F., Olken, Benjamin A., Sautmann, Anja (2020). In Praise of Moderation: Suggestions for the Scope and Use of Pre-Analysis Plans for RCTs in Economics. *NBER Working Paper Series*. https://doi.org/10.3386/w26993
2. Bechhofer, Sean, Roure, David De, Gamble, Matthew, Goble, Carole, Buchan, Iain (2010). Research Objects: Towards Exchange and Reuse of Digital Knowledge. *Nature Precedings*. https://doi.org/10.1038/npre.2010.4626.1
3. Beck, Susanne, Bergenholtz, Carsten, Bogers, Marcel, Brasseur, Tiare-Maria, Conradsen, Marie Louise, Marco, Diletta Di, Distel, Andreas, Dobusch, Leonhard, Dörler, Daniel, Effert, Agnes, Fecher, Benedikt, Filiou, Despoina, Frederiksen, Lars, Gillier, Thomas, Grimpe, Christoph, Gruber, Marc, Haeussler, Carolin, Heigl, Florian, Hoisl, Karin, Hyslop, Katie, Kokshagina, Olga, LaFlamme, Marcel, Lawson, Cornelia, Lifshitz-Assaf, Hila, Lukas, W., Nordberg, Markus, Norn, Maria-Theresa, Poetz, Marion, Ponti, Marisa, Pruschak, Gernot, Priego, Laia Pujol, Radziwon, Agnieszka, Rafner, Janet, Romanova, Gergana Petrova, Ruser, Alexander, Sauermann, Henry, Shah, Sonali, Sherson, Jacob, Suess-Reyes, Julia, Tucci, Christopher L., Tuertscher, Philipp, Vedel, Jane Bjørn, Velden, Theresa, Verganti, Roberto, Wareham, Jonathan, Wiggins, Andrea, Xu, Sunny Mosangzi (2020). The Open Innovation in Science research field: a collaborative conceptualisation approach. *Industry and Innovation*. https://doi.org/10.1080/13662716.2020.1792274
4. Benjamini, Yoav, Yekutieli, Daniel (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*. https://doi.org/10.1214/aos/1013699998
5. Boiko, Daniil A., MacKnight, Robert, Kline, Ben, Gomes, Gabriel dos Passos (2023). Autonomous chemical research with large language models. *Nature*. https://doi.org/10.1038/s41586-023-06792-0
6. Borusyak, Kirill, Jaravel, Xavier, Spiess, Jann (2024). Revisiting Event-Study Designs: Robust and Efficient Estimation. *Review of Economic Studies*. https://doi.org/10.1093/restud/rdae007
7. Bran, Andres M., Cox, Sam, Schilter, Oliver, Baldassari, Carlo, White, Andrew Dickson, Schwaller, Philippe (2024). Augmenting large language models with chemistry tools. *Nature Machine Intelligence*. https://doi.org/10.1038/s42256-024-00832-8

8. Carroll, Stephanie Russo, Garba, Ibrahim, Rodríguez, Óscar Luis Figueroa, Holbrook, Jarita, Lovett, Raymond, Materechera, S. A., Parsons, M. A., Raseroka, Kay, Rodriguez-Lonebear, Desi, Rowe, Robyn, Sara, Rodrigo, Walker, Jennifer, Anderson, Jane, Hudson, Māui (2020). The CARE Principles for Indigenous Data Governance. *Data Science Journal*. https://doi.org/10.5334/dsj-2020-043
9. Chang, Andrew C., Li, Phillip (2015). Is Economics Research Replicable? Sixty Published Papers from Thirteen Journals Say ``Usually Not''. *Finance and Economics Discussion Series*. https://doi.org/10.17016/feds.2015.083
10. Charness, Gary, Jabarian, Brian, List, John A. (2023). Generation Next: Experimentation with AI. *NBER Working Paper Series*. https://doi.org/10.3386/w31679
11. Chen, Chuke, Luo, Biao, Li, Nan, Wang, Boxiang, Yang, Hang, Guo, Jing, Xu, Ming (2025). Spec-Driven AI for Science: The ARIA Framework for Automated and Reproducible Data Analysis. *arXiv*. https://arxiv.org/abs/2510.11143
12. Christensen, Garret, Miguel, Edward (2018). Transparency, Reproducibility, and the Credibility of Economics Research. *Journal of Economic Literature*. https://doi.org/10.1257/jel.20171350
13. Christensen, Garret, Dafoe, Allan, Miguel, Edward, Moore, Don A., Rose, Andrew K. (2019). A study of the impact of data sharing on article citations using journal policies as a natural experiment. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0225883
14. Duvendack, Maren, Palmer-Jones, Richard, Reed, W. Robert (2017). What Is Meant by ``Replication'' and Why Does It Encounter Resistance in Economics?. *American Economic Review*. https://doi.org/10.1257/aer.p20171031
15. Forkel, Robert, List, Johann-Mattis, Greenhill, Simon J., Rzymski, Christoph, Bank, Sebastian, Cysouw, Michael, Hammarström, Harald, Haspelmath, Martín, Kaiping, Gereon A., Gray, Russell D. (2018). Cross-Linguistic Data Formats, advancing data sharing and re-use in comparative linguistics. *Scientific Data*. https://doi.org/10.1038/sdata.2018.205
16. Gao, Chen, Lan, Xiaochong, Li, Nian, Yuan, Yuan, Ding, Jingtao, Zhou, Zhilun, Xu, Fengli, Li, Yong (2024). Large language models empowered agent-based modeling and simulation: a survey and perspectives. *Humanities and Social Sciences Communications*. https://doi.org/10.1057/s41599-024-03611-3
17. German Journal of Agricultural Economics (2023). The p-Value Debate and Statistical (Mal)practice – Implications for the Agricultural and Food Economics Community. *German Journal of Agricultural Economics*. https://doi.org/10.30430/gjae.2023.0231
18. Giofrè, David, Cumming, Geoff, Fresc, Luca, Boedker, Ingrid, Tressoldi, Patrizio (2017). The influence of journal submission guidelines on authors' reporting of statistics and use of open research practices. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0175583
19. Hagger, Martin S., Chatzisarantis, Nikos L. D., Alberts, H., Anggono, Calvin Octavianus, Batailler, Cédric, Birt, Angela R., Brand, Ralf, Brand, Muriele, Brewer, Gayle, Bruyneel, S., Calvillo, Dustin P., Campbell, Wenonah, Cannon, Peter Robert, Carlucci, M., Carruth, Nicholas P., Cheung, Tracy, Crowell, Adrienne, Ridder, Denise de, Dewitte, Siegfried, Elson, Malte, Evans, Jacqueline R., Fay, Anne L., Fennis, Bob M., Finley, Anna J, Francis, Zoë, Heise, Elke, Hoemann, Harry W., Inzlicht, Michael, Koole, Sander L., Koppel, Lina, Kroese, Floor M., Lange, Florian, Lau, Karen, Lynch, Patricia A., Martijn, Carolien, Merckelbach, Harald, Mills, Neil P., Michirev, Alexej, Miyake, Akira, Mosser, A., Muise, Megan D., Müller, Dominique, Muzi, M Gallinella, Nalis, Dario, Nurwanti, Ratri, Otgaar, Henry, Philipp, Michael, Primoceri, Pierpaolo, Rentzsch, Katrin, Ringos, Lara Jade, Schlinkert, Caroline, Schmeichel, Brandon J., Schoch, Sarah F., Schrama, Maarten, Schütz, Astrid, Stamos, Angelos, Tinghög, Gustav, Ullrich, Johannes, vanDellen, Michelle R., Wimbarti, Supra, Wolff, Wanja, Yusainy, Cleoputri,

Zerhouni, Oulmann, Zwienenberg, M. (2016). A Multilab Preregistered Replication of the Ego-Depletion Effect. *Perspectives on Psychological Science*. https://doi.org/10.1177/1745691616652873
20. Ioannidis, John P. A. (2005). Why Most Published Research Findings Are False. *PLoS Medicine*. https://doi.org/10.1371/journal.pmed.0020124
21. Kerr, Norbert L. (1998). HARKing: Hypothesizing After the Results are Known. *Personality and Social Psychology Review*. https://doi.org/10.1207/s15327957pspr0203_4
22. Kurtzer, Gregory M., Sochat, Vanessa, Bauer, Michael W. (2017). Singularity: Scientific containers for mobility of compute. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0177459
23. List, John A., Shaikh, Azeem M., Xu, Yang (2016). Multiple Hypothesis Testing in Experimental Economics. *National Bureau of Economic Research*. https://doi.org/10.3386/w21875
24. Lu, Chris, Lu, Cong, Lange, Robert Tjarko, Foerster, Jakob, Clune, Jeff, Ha, David (2024). The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv*. https://arxiv.org/abs/2408.06292
25. Menkveld, Albert J., Dreber, Anna, Holzmeister, Felix, Huber, Juergen, Johannesson, Magnus, Kirchler, Michael, others (2024). Nonstandard Errors. *Journal of Finance*. https://doi.org/10.1111/jofi.13337
26. Miguel, Edward (2021). Evidence on Research Transparency in Economics. *Journal of Economic Perspectives*. https://doi.org/10.1257/jep.35.3.193
27. Moher, David, Bouter, L.M., Kleinert, Sabine, Glasziou, Paul, Sham, MH, Barbour, Virginia, Coriat, Anne-Marie, Foeger, Nicole, Dirnagl, Ulrich (2020). The Hong Kong Principles for assessing researchers: Fostering research integrity. *PLoS Biology*. https://doi.org/10.1371/journal.pbio.3000737
28. Munafò, Marcus R., Nosek, Brian A., Bishop, Dorothy, Button, Katherine S., Chambers, Chris, Sert, Nathalie Percie du, Simonsohn, Uri, Wagenmakers, Eric-Jan, Ware, Jennifer J., Ioannidis, John P. A. (2017). A manifesto for reproducible science.*Nature Human Behaviour*. https://doi.org/10.1038/s41562-016-0021
29. Ofosu, George Kwaku, Posner, Daniel (2020). Do Pre-analysis Plans Hamper Publication?. *AEA Papers and Proceedings*. https://doi.org/10.1257/pandp.20201079
30. Olken, Benjamin (2015). Promises and Perils of Pre-Analysis Plans. *The Journal of Economic Perspectives*. https://doi.org/10.1257/jep.29.3.61
31. Open Science Collaboration (2015). Estimating the reproducibility of psychological science. *Science*. https://doi.org/10.1126/science.aac4716
32. Ramos, Mayk Caldas, Collison, Christopher J., White, Andrew Dickson (2024). A review of large language models and autonomous agents in chemistry. *Chemical Science*. https://doi.org/10.1039/d4sc03921a
33. Romano, Joseph P., Wolf, Michael (2005). Stepwise Multiple Testing as Formalized Data Snooping. *Econometrica*. https://doi.org/10.1111/j.1468-0262.2005.00615.x
34. Schmidgall, Samuel, Su, Yusheng, Wang, Ze, Sun, Ximeng, Wu, Jialian, Yu, Xiaodong, Liu, Jiang, Moor, Michael, Liu, Zicheng, Barsoum, Emad (2025). Agent Laboratory: Using LLM Agents as Research Assistants. *arXiv*. https://arxiv.org/abs/2501.04227
35. Simmons, Joseph P., Nelson, Leif D., Simonsohn, Uri (2011). False-Positive Psychology. *Psychological Science*. https://doi.org/10.1177/0956797611417632
36. Simonsohn, Uri, Simmons, Joseph P., Nelson, Leif D. (2020). Specification curve analysis. *Nature Human Behaviour*. https://doi.org/10.1038/s41562-020-0912-z

37. Soiland-Reyes, Stian, Sefton, Peter, Crosas, Mercè, Castro, Leyla Jael, Coppens, Frederik, Fernández, José M., Garijo, Daniel, Grüning, Björn, Rosa, M., Leo, Simone, Carragáin, Eoghan Ó, Portier, Marc, Trisovic, A., Community, RO-Crate, Groth, Paul, Goble, Carole (2022). Packaging research artefacts with RO-Crate. *Data Science*. https://doi.org/10.3233/ds-210053
38. Steegen, Sara, Tuerlinckx, Francis, Gelman, Andrew, Vanpaemel, Wolf (2016). Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*. https://doi.org/10.1177/1745691616658637
39. Vilhuber, Lars (2020). Reproducibility and Replicability in Economics. *Harvard Data Science Review*. https://doi.org/10.1162/99608f92.4f6b9e67
40. Wallis, Jillian C., Rolando, Elizabeth, Borgman, Christine L. (2013). If We Share Data, Will Anyone Use Them? Data Sharing and Reuse in the Long Tail of Science and Technology. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0067332
41. White, Halbert (2000). A Reality Check for Data Snooping. *Econometrica*. https://doi.org/10.1111/1468-0262.00152
42. Wilkinson, Mark D., Dumontier, Michel, Aalbersberg, IJsbrand Jan, Appleton, Gabrielle, Axton, Myles, Baak, Arie, Blomberg, Niklas, Boiten, Jan-Willem, Santos, Luiz Olavo Bonino da Silva, Bourne, Philip E., Bouwman, Jildau, Brookes, Anthony J., Clark, Tim W., Crosas, Mercè, Dillo, Ingrid, Dumon, Olivier, Edmunds, Scott, Evelo, Chris T., Finkers, Richard, González-Beltrán, Alejandra, Gray, Alasdair J. G., Groth, Paul, Goble, Carole, Grethe, Jeffrey S., Heringa, Jaap, Hoen, Peter A.C. 't, Hooft, Rob, Kuhn, Tobias, Kok, Ruben, Kok, Joost N., Lusher, Scott J., Martone, Maryann E., Mons, Albert, Packer, Abel L., Persson, Bengt, Rocca-Serra, Philippe, Roos, Marco, Schaik, René van, Sansone, Susanna-Assunta, Schultes, Erik, Sengstag, Thierry, Slater, Ted, Strawn, George, Swertz, Morris A., Thompson, Mark, Lei, Johan van der, Mulligen, Erik M. van, Velterop, Jan, Waagmeester, Andra, Wittenburg, Peter, Wolstencroft, Katherine, Zhao, Jun, Mons, Barend (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*. https://doi.org/10.1038/sdata.2016.18
43. Wilkinson, Sean R., Aloqalaa, Meznah, Belhajjame, Khalid, Crusoe, Michael R., Kinoshita, Bruno P., Gadelha, Luiz, Garijo, Daniel, Gustafsson, Ove, Juty, Nick, Kanwal, Sehrish, Khan, Farah Zaib, Köster, Johannes, Peters, Karsten, Pouchard, Line, Rannow, Randy K., Soiland-Reyes, Stian, Soranzo, Nicola, Sufi, Shoaib, Sun, Ziheng, Vilne, Baiba, Wouters, Merridee A., Yuen, Denis, Goble, Carole (2025). Applying the FAIR Principles to computational workflows. *Scientific Data*. https://doi.org/10.1038/s41597-025-04451-9
44. Wojarnik, Grzegorz (2026). FSAIM: Spec-driven AI as a Governance Layer for Empirical Research – A Position Note. *SSRN*. https://doi.org/10.2139/ssrn.6853581